Final Lab Product Evaluation Worksheet

Instructions: Review the Product Concept Document. Respond to the following questions.

Part I: List at Least Five Skills required by the AI Product Manager

- Review the AI Skills Checklist
- Determine the top five skills required for the AI Product Manager to be successful. Justify your selections

Top Skills Required	Rationale
AI Literacy & Technical Understanding	To make informed decisions about model capabilities, data needs, limitations, and feasibility of features like real-time routing and prediction.
Data Analysis & Data Management Skills	The product relies heavily on large-scale real-time traffic data, requiring strong skills in validating data quality, defining data pipelines, and understanding data dependencies.
User-Centered Design & Customer Empathy	Ensures the product meets commuter needs, is easy to use, and provides accurate, helpful recommendations that drive adoption.
Cross-Functional Collaboration	Necessary to coordinate with engineers, data scientists, DOT stakeholders, marketing, and leadership to align requirements and timelines.
Risk Management & Ethical AI Awareness	Critical to address privacy, bias, compliance, and accuracy risks—especially since the system interacts with public infrastructure and user behavior.

Part II: Identify Five Challenges or Risks that the Team May Encounter

- Review readings and videos in the AI Methods, Challenges, and Opportunities Lesson
- Identify at least five challenges or risks, and outline a mitigation response for each

Five Challenges or Risks	Planned Mitigation Responses
Data Quality and Availability	Use diverse, reliable data sources; implement automated data validation; collaborate with DOT for data-sharing agreements.
Model Accuracy & Real-Time Performance	Conduct extensive model testing; deploy monitoring dashboards; enable human override for critical alerts.

Privacy, Security & Compliance Risks	Implement privacy-by-design, encryption, anonymization; comply with federal/state transportation and data laws.
Integration with DOT & Legacy Systems	Plan API integration early; run compatibility assessments; pilot integration in limited areas first.
User Adoption & Trust Issues	Provide clear value through accurate routing; ensure intuitive UX; launch educational campaigns showing benefits.

Part III: Map Planning and Deployment Activities to the Four-Step AI Product Deployment Process

- Review the Video: Stages of AI Product Development
- Map at least three steps to effectively plan and launch this new product in each stage

Product Development Stages	Required Steps/Activities
Ideation and Innovation Stage Requirements	Define the problem statement and success criteria (travel time reduction, real-time accuracy, user satisfaction).
	Conduct competitor analysis (e.g., Project Green Light).
	Identify key stakeholders (commuters, DOT, municipal agencies).
	Draft high-level feature requirements and AI use-cases.
Data Management Requirements	Determine required datasets: traffic sensors, GPS data, weather, DOT feeds.
	Build the data ingestion pipeline and architecture.
	Set data quality rules and privacy safeguards.
	Establish continuous real-time data update systems.
Research and Development Requirements	Experiment with AI/ML models for route optimization and event detection.
	Develop prototypes and run simulations using historical traffic data.
	Perform user testing on routing accuracy and notification relevance.
	Refine models and integrate them with the mobile application.
Deployment Requirements	Conduct alpha and beta pilot testing in limited city zones.
	Establish real-time monitoring dashboards for system performance.

	Roll out the app city-wide; train DOT operators to use system insights.
	Implement continuous updates based on live data and feedback.

Part IV: Determine How to Effectively Market this Product

- Review the Video: How to Commercialize AI Products.
- Share at least three strategies you will use to convince Pragmatists and Conservatives to use this new service

Strategies to Convince Pragmatists and Conservatives
Showcase proven results using real pilot data Pragmatists need evidence: reduced travel times, improved traffic flow, and user satisfaction metrics.
Leverage endorsements from trusted institutions (DOT, city government) Conservatives rely on authority and validation from established organizations.
Highlight reliability, security, and regulatory compliance These groups prioritize safety, stability, and minimizing risk.
Provide clear onboarding, tutorials, and support Reduces fear of adopting new technology and builds trust.